# Case_in_HPC_Assign2

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March 2026

## 1 Introduction

This experiment implemented the GMRES algorithm to solve linear equations.

$$Ax = b.$$

This experiment implemented three different versions of the GMRES algorithm and conducted tests and comparisons on both small-scale and large-scale matrices within the SuiteSparse matrix library.

## 2 Algorithm Implementation

This experiment implemented the following three GMRES methods:

### 2.1 Basic GMRES

In the basic version, at each iteration step, an orthogonal basis is constructed through the Arnoldi process, and the least squares problem is directly solved:

$$\min \|\beta e_1 - H_m y\|.$$

Then, an approximate solution is constructed:

$$x_m = x_0 + V_m y.$$

Among them:

- $V_m$ is an orthogonal basis matrix
- $H_m$ is an upper Hessenberg matrix

### 2.2 Givens Rotation GMRES (givens)

This version uses successive Givens rotations to perform QR decomposition on the Hessenberg matrix, avoiding the repeated solution of the least squares problem, thereby improving computational efficiency.

### 2.3 Residual Monitoring GMRES (monitor)

This version uses the residual estimates obtained during the Givens rotation process to determine convergence, rather than constructing an approximate solution at each step. Only after meeting the convergence criteria is the final solution calculated, thereby reducing computational costs.

## 3 Numerical Experiment

In the experiment, the tolerance was set as

$$\text{tol} = 10^{-8},$$

The maximum number of iterations is

$$\text{maxit} = 200.$$

### 3.1 Small-scale Matrix (cdde6)

Select the matrix cdde6 from SuiteSparse, which has a dimension of 961. The experimental results are as follows:

- Basic residual $= 8.734 \times 10^{-9}$
- Givens residual $= 8.734 \times 10^{-9}$
- Monitor residual $= 8.734 \times 10^{-9}$
- Number of convergence iterations: 78

It can be seen that all three methods have successfully converged within the specified tolerance range, and the results are completely identical.

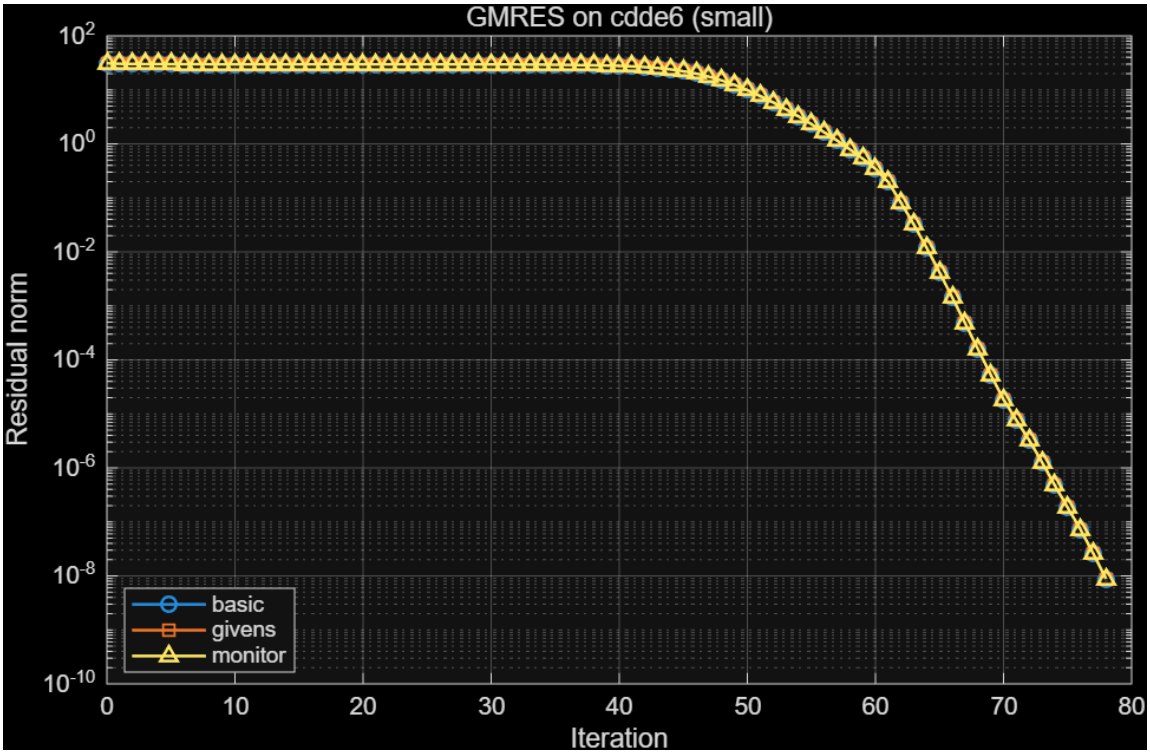


Figure 1: Convergence curve of GMRES on the cdde6 matrix

As can be seen from the graph, the residuals changed relatively slowly in the early stage. They began to drop rapidly after approximately the 50th iteration and finally reached the set tolerance. The curves of the three methods almost completely overlap, indicating that they are consistent in terms of numerical results.

### 3.2 Large-scale Matrix (parabolic_fem)

Select the large-scale matrix parabolic_fem from SuiteSparse, with a dimension of 525825. The experimental results are as follows:

- Basic residual = 1.499
- Givens residual = 1.499
- Monitor residual = 1.499
- Number of iterations: 200 (reached the maximum number of iterations)

It can be observed that within the maximum iteration count of 200, GMRES failed to converge to the specified tolerance, but the residual showed a continuous downward trend.

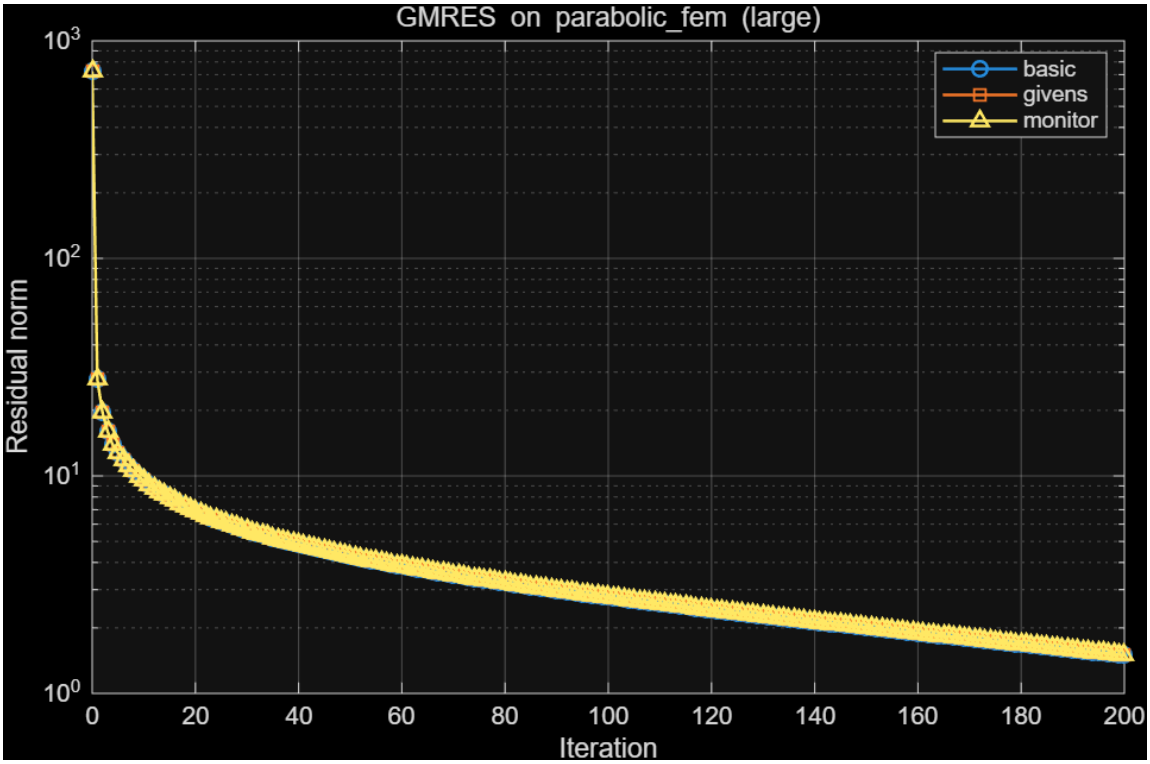


Figure 2: Convergence curve of GMRES on the parabolic_fem matrix

# 4 Result Analysis

The following conclusions can be drawn from the experiment:

- The three implementations of GMRES produced the same residual results in all tests, which indicates that the three algorithms are numerically equivalent and also verifies the correctness of the code implementation.
- For the cdde6 matrix, GMRES successfully converged within 78 steps. This indicates that the spectral properties of this matrix are favorable, making it suitable for solution by GMRES. From the convergence curve, it can be observed that the residuals decrease relatively slowly in the early stage and then drop rapidly in the later stage. This phenomenon is quite common in the GMRES method.
- For the parabolic_fem matrix, the convergence condition was not achieved within 200 iterations. This indicates that the problem is rather challenging and may have the following characteristics:
  - The matrix condition number is large.
  - The distribution of eigenvalues is not conducive to the convergence of GMRES.

  - No preconditioning technique was used.

  Nevertheless, the residuals still show a stable downward trend, indicating that the algorithm is gradually approaching the solution.

Although the convergence results of the three implementation methods are the same, their calculation methods are different:

- The basic version constructs solutions at each step, resulting in a large amount of computation.
- The givens version improves efficiency through QR updates.
- The monitor version avoids frequently constructing solutions and is more efficient.

Therefore, in practical applications, the latter two methods have greater advantages.

# 5 Conclusion

This experiment successfully implemented the three forms of the GMRES algorithm and verified its effectiveness on matrices of different sizes. The results show that GMRES performs well in small-scale problems, but in large-scale problems, more iterations or the introduction of preconditioning techniques are required to improve the convergence speed.